

Lecture 1: Machine Learning in 2026 and a Mathematical Review + Bridge

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SoE, Westlake University

Fall 2026



1 Machine Learning in 2026

2 Review: Linear Algebra

- Notation
- Vectors
- Matrices

3 Calculus Recap

4 Review: Probability Theory

- Elements of probability
- Random variables
- Two random variables

What you should be able to explain after today

From examples to a learning problem

1. Identify inputs, targets, parameters, loss, training, and evaluation.
2. Explain how representations turn numbers into task-relevant features.

From modern systems to mathematical review

3. Relate algorithms, data, compute, and systems without assuming unlimited scaling.
4. Review geometry and uncertainty, then bridge gradients and composition to later lectures.

45 min Modern ML
data → models → decisions

45 min Review + bridge
linear algebra review;
two calculus recaps

45 min Probability review
distributions, conditioning, Bayes

Check yourself

Check yourself (rapid check)

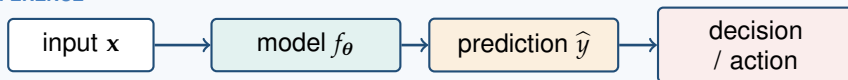
Training loss falls while test error rises. Which part of the pipeline is warning us?

Commit to an answer before advancing the slide.

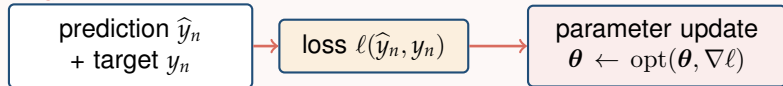
A learning system connects data to decisions

training examples $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$, $\mathbf{x}_n \in \mathcal{X} \subseteq \mathbb{R}^D$, $y_n \in \mathcal{Y}$

INFERENCE



TRAINING



Evaluation is separate: test the trained inference path on held-out data, relevant shifts, and deployment costs.

Check yourself (rapid check) — reveal

Question: Training loss falls while test error rises. Which part of the pipeline is warning us?

Answer: Evaluation exposes a growing generalization gap; lower training loss alone is not the goal.

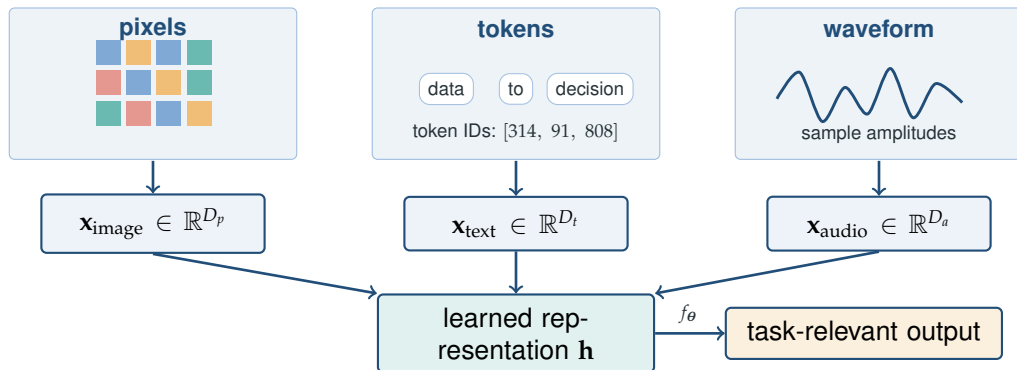
Rapid check

Rapid check

If two inputs are close as raw vectors, must they mean the same thing?

Commit to an answer before advancing the slide.

Inputs are numbers before they become meaning

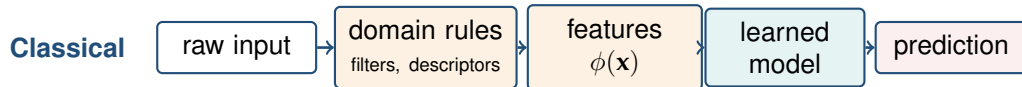


Rapid check — reveal

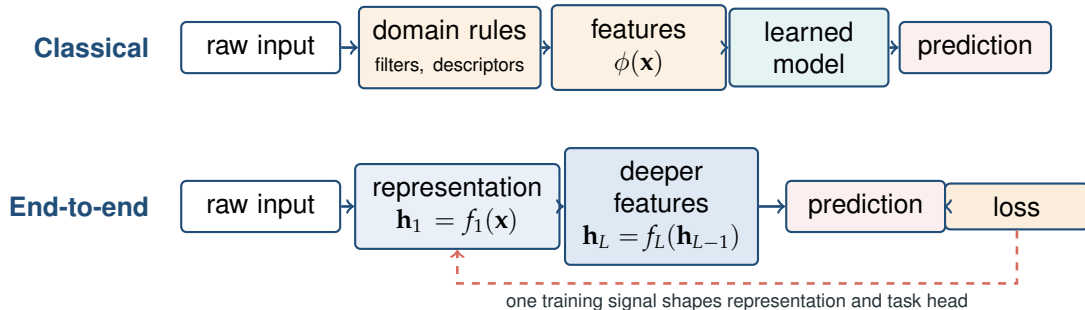
Question: If two inputs are close as raw vectors, must they mean the same thing?

Answer: No. Coordinates carry measurements; raw distance depends on the representation and may not align with semantics.

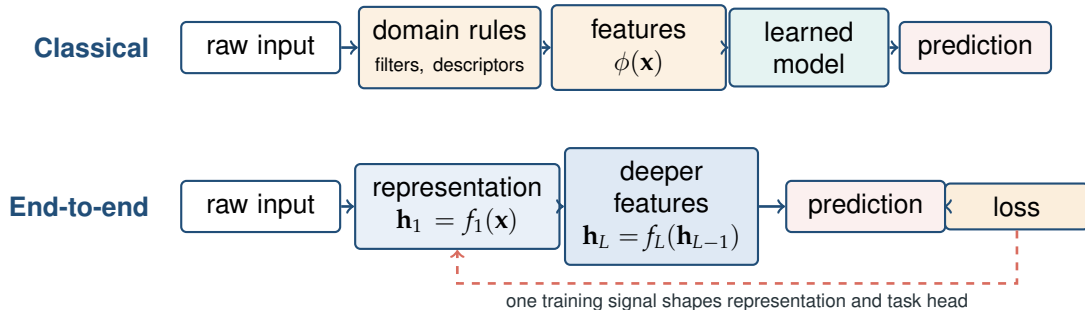
Representation learning reduces hand-designed interfaces



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Representation learning reduces hand-designed interfaces



The interface moves; it does not disappear

Learned features reduce manual feature engineering, while people still choose data, architecture, objectives, constraints, and evaluation.

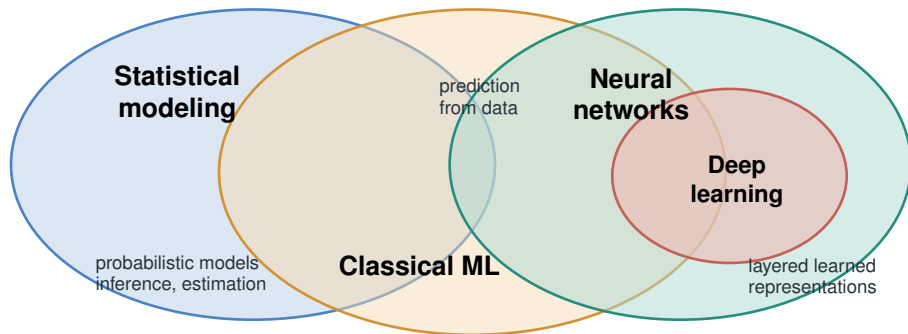
Optional rapid check

Optional rapid check

Is deep learning outside statistical modeling?

Commit to an answer before advancing the slide.

ML traditions overlap

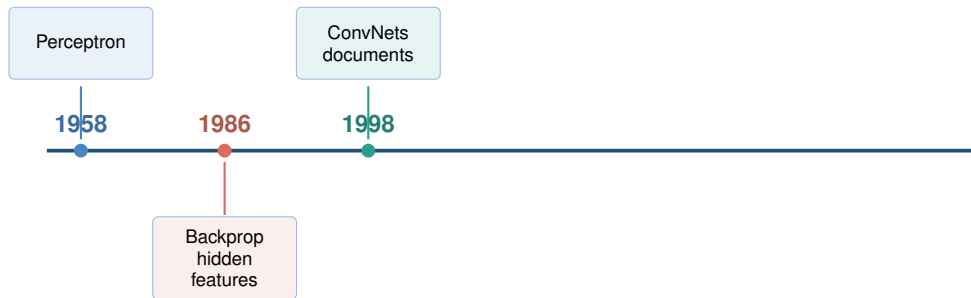


Optional rapid check — reveal

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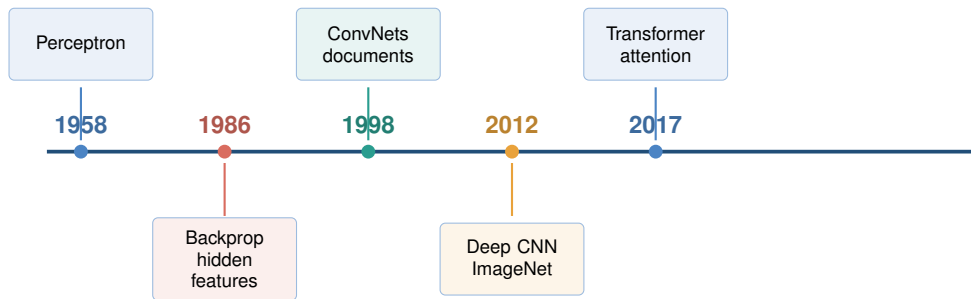
Answer: No. These communities and toolkits overlap rather than form a historical ladder; modern neural systems still use statistical objectives and inference.

The ideas are old; the operating regime changed



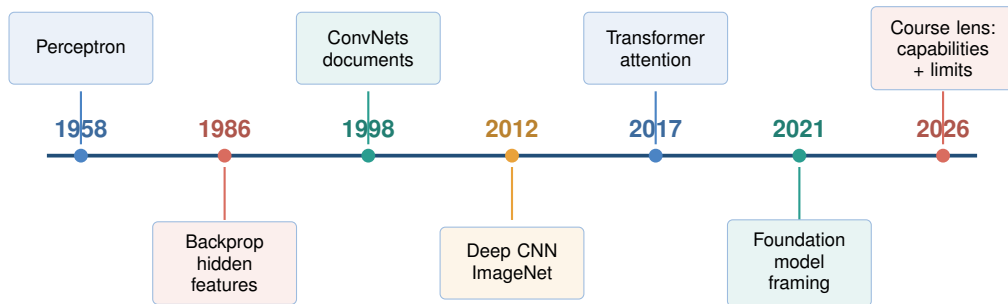
Sources: [S12] Rosenblatt, *Psychological Review*, 1958; [S01] Rumelhart, Hinton, and Williams, *Nature*, 1986; [S02] LeCun et al., *Proc. IEEE*, 1998; [S03] Krizhevsky, Sutskever, and Hinton, NeurIPS, 2012; [S04] Vaswani et al., NeurIPS, 2017; [S05] Bommasani et al., Stanford CRFM, 2021.

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The ideas are old; the operating regime changed



Reusable ingredients accumulated; data scale, parallel compute, software stacks, and broad adaptation changed what could be built together.

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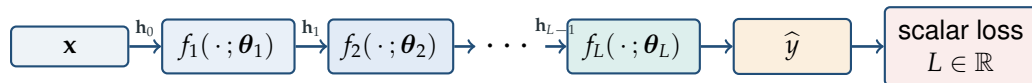
Optional rapid check

Optional rapid check

What information from the forward pass might the backward pass need?

Commit to an answer before advancing the slide.

A deep model is a composition of parameterized maps



$$f_{\boldsymbol{\theta}} = f_L \circ \dots \circ f_2 \circ f_1, \quad \boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_L).$$

$$\nabla_{\mathbf{h}_{\ell-1}} L = \mathbf{J}_{f_{\ell}}(\mathbf{h}_{\ell-1})^\top \nabla_{\mathbf{h}_{\ell}} L, \quad \mathbf{J}_{f_{\ell}} \in \mathbb{R}^{d_{\ell} \times d_{\ell-1}}.$$

Optional rapid check — reveal

Question: What information from the forward pass might the backward pass need?

Answer: Intermediate activations and local derivatives—or recomputation of them—connect the local chain-rule factors while shapes match at every arrow.

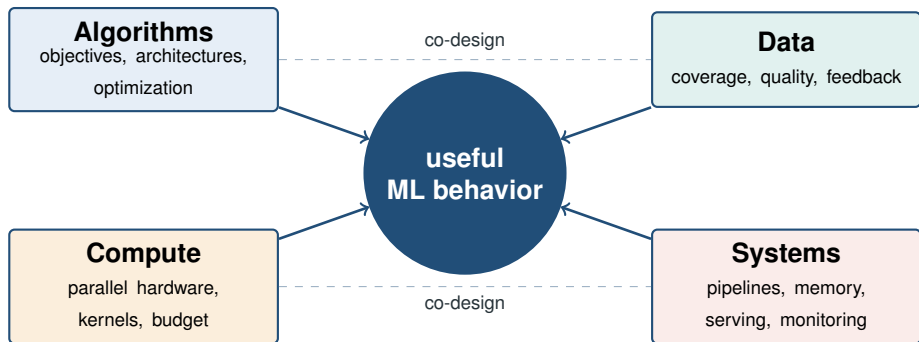
Think–Pair–Share

Think–Pair–Share

Rank the four factors—or choose the single factor that best explains modern progress.

Commit to an answer before advancing the slide.

Progress combines algorithms, data, compute, and systems



Think–Pair–Share — reveal

Question: Rank the four factors—or choose the single factor that best explains modern progress.

Answer: No single factor is sufficient. Progress comes from their interaction, and the bottleneck shifts by regime; more of one cannot automatically repair poor coverage, the wrong objective, an unreliable pipeline, or an infeasible budget.

Perception and generation now share multimodal models



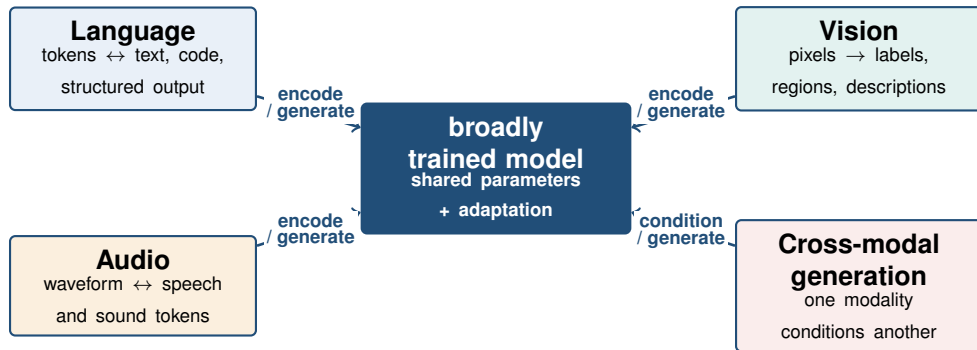
Sources: [S05] Bommasani et al., Stanford CRFM, 2021; [S06] AI Index Steering Committee, Stanford HAI, 2026.

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Perception and generation now share multimodal models



Shared interfaces enable many tasks; they do not imply equal competence across modalities, benchmarks, or deployment settings.

Sources: [S05] Bommasani et al., Stanford CRFM, 2021; [S06] AI Index Steering Committee, Stanford HAI, 2026.

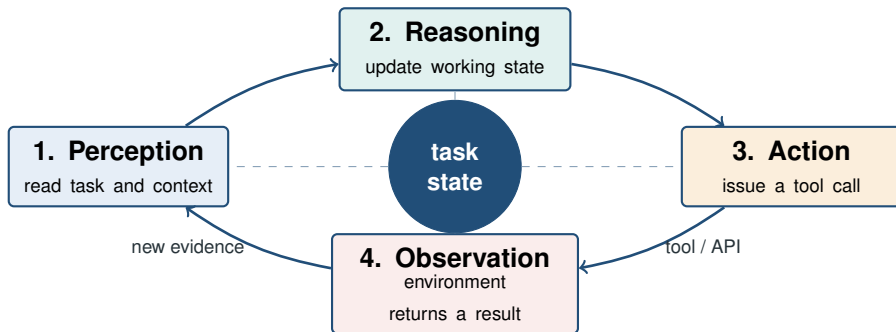
Pause & Predict

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What turns a chatbot into an agent: a larger model, or a feedback loop with actions and observations?

Commit to an answer before advancing the slide.

Agents couple models to tools and environments



Pause & Predict — reveal

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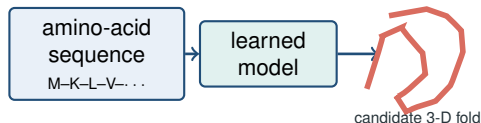
Answer: Agency here comes from coupling the model to tools and environment feedback; scale alone does not define it. ReAct interleaves reasoning, actions, and observations, while permissions, stopping, verification, and oversight remain system choices.

Sources: [S10] Yao et al., ICLR, 2023.

Machine learning is becoming a scientific instrument

Example: protein structure prediction

In CASP14, AlphaFold achieved accuracy competitive with experimental structures in a majority of cases and substantially exceeded other entered methods.

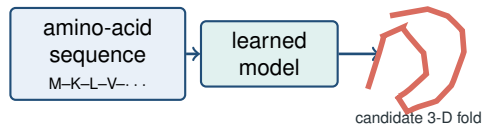


Sources: [S09] Jumper et al., *Nature*, 2021.

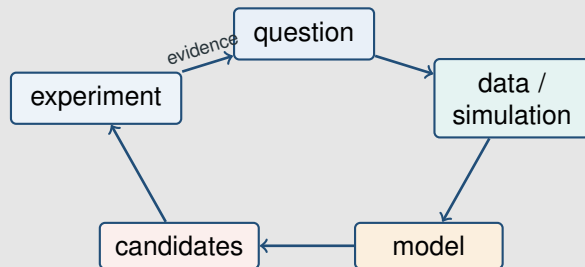
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General discovery loop



ML can prioritize hypotheses or infer latent structure; scientific claims still depend on domain assumptions, uncertainty, and empirical validation.

Sources: [S09] Jumper et al., *Nature*, 2021.

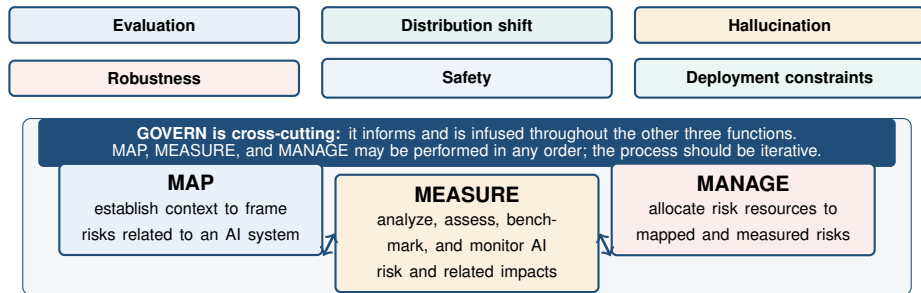
Think–Pair–Share

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A model exceeds a benchmark's human baseline. Is it deployment-ready?

Commit to an answer before advancing the slide.

Benchmark gains do not eliminate reliability gaps



Think–Pair–Share — reveal

Question: A model exceeds a benchmark's human baseline. Is it deployment-ready?

Answer: Not necessarily. Benchmark scope, distribution shift, calibration, robustness, safety, and operating context still matter.

Sources: [S06] AI Index Steering Committee, Stanford HAI, 2026; [S11] Tabassi and NIST, AI RMF 1.0, 2023.

Pause & Predict

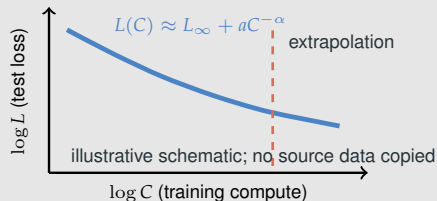
Pause & Predict

With a fixed compute budget, should we keep increasing parameters while holding data fixed?

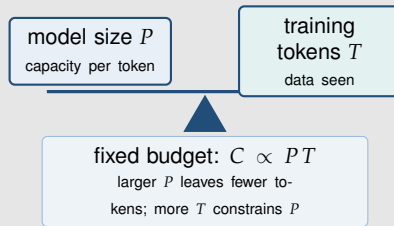
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Scaling laws describe regimes, not guarantees

Empirical fit in a measured regime



Finite compute forces a tradeoff



- Power laws summarized loss trends in the language-model regimes studied by Kaplan et al.
- Hoffmann et al. found near-equal model/token scaling under their compute-optimal setup across more than 400 runs.
- A regime can change with data quality, architecture, objective, or evaluation; extrapolation is a hypothesis.

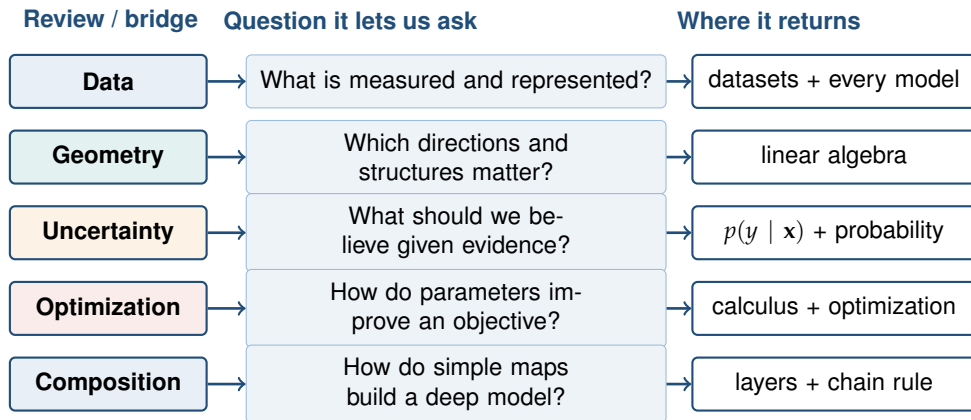
Pause & Predict — reveal

Question: With a fixed compute budget, should we keep increasing parameters while holding data fixed?

Answer: No. Compute-optimal training balances model size and training data, and the empirical relation is regime-dependent.

Sources: [S07] Kaplan et al., 2020; [S08] Hoffmann et al., 2022.

Next: review the reusable mathematical language



We review familiar mathematics briskly and add only the two-frame calculus bridge needed later.

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Notation

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- **Sets** by uppercase calligraphic letters (e.g., $\mathcal{S}$).

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Vector spaces

Vector spaces

Note 1

We focus on the **field of real numbers** ($\mathbb{R}$) but most of the results can be *generalized* to **the field of complex numbers** ($\mathbb{C}$).

Vector spaces

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Recall

Can you name the addition, scalar-multiplication, distributive, and identity axioms before they are revealed?

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commutative under addition

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- 2 $(x + y) + z = x + (y + z), \forall x, y, z \in \mathcal{V}$

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0 being additive identity

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- 3 $0 + x = x, \forall x \in \mathcal{V}$
- 4 $\forall x \in \mathcal{V} \quad \exists (-x) \in \mathcal{V} \text{ such that } x + (-x) = 0$

commutative under addition

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0 being additive identity

$-x$ being additive inverse

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- 2 $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z}), \quad \forall \mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathcal{V}$
- 3 $\mathbf{0} + \mathbf{x} = \mathbf{x}, \quad \forall \mathbf{x} \in \mathcal{V}$
- 4 $\forall \mathbf{x} \in \mathcal{V} \quad \exists (-\mathbf{x}) \in \mathcal{V} \text{ such that } \mathbf{x} + (-\mathbf{x}) = \mathbf{0}$
- 5 $(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x}), \quad \alpha, \beta \in \mathbb{R} \quad \forall \mathbf{x} \in \mathcal{V}$

commutative under addition

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$-\mathbf{x}$ being additive inverse

associative under scalar multiplication

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- 3 $0 + x = x, \quad \forall x \in \mathcal{V}$ 0 being additive identity
- 4 $\forall x \in \mathcal{V} \quad \exists (-x) \in \mathcal{V}$ such that $x + (-x) = 0$ $-x$ being additive inverse
- 5 $(\alpha\beta)x = \alpha(\beta x), \quad \alpha, \beta \in \mathbb{R} \quad \forall x \in \mathcal{V}$ associative under scalar multiplication
- 6 $\alpha(x + y) = \alpha x + \alpha y, \quad \forall \alpha \in \mathbb{R} \quad \forall x, y \in \mathcal{V}$ distributive

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- 4 $\forall \mathbf{x} \in \mathcal{V} \quad \exists (-\mathbf{x}) \in \mathcal{V} \text{ such that } \mathbf{x} + (-\mathbf{x}) = \mathbf{0}$ $-\mathbf{x}$ being additive inverse
- 5 $(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x}), \quad \alpha, \beta \in \mathbb{R} \quad \forall \mathbf{x} \in \mathcal{V}$ associative under scalar multiplication
- 6 $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}, \quad \forall \alpha \in \mathbb{R} \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{V}$ distributive
- 7 $(\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}, \quad \forall \alpha, \beta \in \mathbb{R} \quad \forall \mathbf{x} \in \mathcal{V}$ distributive

Vector spaces

Note 1

We focus on the **field of real numbers** ($\mathbb{R}$) but most of the results can be **generalized** to the **field of complex numbers** ($\mathbb{C}$).

A vector space or *linear space* (over the field $\mathbb{R}$) consists of

- (a) a **set** of vectors $\mathcal{V}$
- (b) an **addition** operation: $\mathcal{V} \times \mathcal{V} \rightarrow \mathcal{V}$
- (c) a **scalar multiplication** operation: $\mathbb{R} \times \mathcal{V} \rightarrow \mathcal{V}$
- (d) a **distinguished** element $0 \in \mathcal{V}$

and satisfies the following properties:

- 1 $x + y = y + x, \quad x, y \in \mathcal{V}$ commutative under addition
- 2 $(x + y) + z = x + (y + z), \quad \forall x, y, z \in \mathcal{V}$ associative under addition
- 3 $0 + x = x, \quad \forall x \in \mathcal{V}$ 0 being additive identity
- 4 $\forall x \in \mathcal{V} \quad \exists (-x) \in \mathcal{V}$ such that $x + (-x) = 0$ $-x$ being additive inverse
- 5 $(\alpha\beta)x = \alpha(\beta x), \quad \alpha, \beta \in \mathbb{R} \quad \forall x \in \mathcal{V}$ associative under scalar multiplication
- 6 $\alpha(x + y) = \alpha x + \alpha y, \quad \forall \alpha \in \mathbb{R} \quad \forall x, y \in \mathcal{V}$ distributive
- 7 $(\alpha + \beta)x = \alpha x + \beta x, \quad \forall \alpha, \beta \in \mathbb{R} \quad \forall x \in \mathcal{V}$ distributive
- 8 $1x = x, \quad \forall x \in \mathcal{V}$ Scalar 1 being multiplicative identity

Vector spaces contd.

Example 2 (Vector space)

- 1 $\mathcal{V}_1 = \{\mathbf{0}\}$ for $\mathbf{0} \in \mathbb{R}^p$
- 2 $\mathcal{V}_2 = \mathbb{R}^p$
- 3 $\mathcal{V}_3 = \left\{ \sum_{i=1}^k \alpha_i \mathbf{x}_i : \alpha_i \in \mathbb{R} \right\}$ for fixed $\mathbf{x}_i \in \mathbb{R}^p$

Vector spaces contd.

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Definition 3 (Subspace)

Vector spaces contd.

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Definition 3 (Subspace)

A **subspace** $\mathcal{S} \subseteq \mathcal{V}$ contains $\mathbf{0}$ and is closed under linear combinations: if $\mathbf{x}, \mathbf{y} \in \mathcal{S}$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha\mathbf{x} + \beta\mathbf{y} \in \mathcal{S}$.

Vector spaces contd.

Example 2 (Vector space)

- 1 $\mathcal{V}_1 = \{\mathbf{0}\}$ for $\mathbf{0} \in \mathbb{R}^p$
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Example 4 (Subspace)

$\mathcal{V}_1$, $\mathcal{V}_2$, and $\mathcal{V}_3$ in the example above are subspaces of $\mathbb{R}^p$.

Vector spaces contd.

Definition 5 (Span)

The **span** of a set of vectors, $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, is the set of all possible **linear combinations** of these vectors; i.e.,

$$\text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\} = \{\alpha_1 \mathbf{x}_1 + \dots + \alpha_k \mathbf{x}_k \mid \alpha_1, \dots, \alpha_k \in \mathbb{R}\}. \quad (1)$$

Vector spaces contd.

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Definition 6 (Linear independence)

A set of vectors, $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, is **linearly independent** if

Vector spaces contd.

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Definition 6 (Linear independence)

A set of vectors, $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, is **linearly independent** if

$$\alpha_1 \mathbf{x}_1 + \dots + \alpha_k \mathbf{x}_k = \mathbf{0} \Rightarrow \alpha_1 = \alpha_2 = \dots = \alpha_k = 0. \quad (2)$$

Vector spaces contd.

Definition 7 (Basis)

The **basis** of a vector space, $\mathcal{V}$, is a set of vectors $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ that satisfy

- 1 $\mathcal{V} = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$,
- 2 $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ are linearly independent.

Vector spaces contd.

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Definition 8 (Dimension)

Every basis of a finite-dimensional vector space $\mathcal{V}$ has the same number of vectors. This common number is the **dimension** of $\mathcal{V}$, denoted $\dim(\mathcal{V})$.

Vector Norms

Definition 9 (Vector norm)

A norm of a vector in $\mathbb{R}^p$ is a function $\|\cdot\| : \mathbb{R}^p \rightarrow \mathbb{R}$ such that for all vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^p$ and scalar $\lambda \in \mathbb{R}$

Recall

Which four properties distinguish a norm from an arbitrary size function?

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(a) $\|\mathbf{x}\| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^p$

non-negativity

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definitiveness

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(c) $\|\lambda \mathbf{x}\| = |\lambda| \|\mathbf{x}\|$

Homogeneity

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triangle inequality

- There is a family of ℓ_q -norms parameterized by $q \in [1, \infty]$;
- For $1 \leq q < \infty$, $\|\mathbf{x}\|_q := \left(\sum_{i=1}^p |x_i|^q\right)^{1/q}$; the endpoint is defined separately by $\|\mathbf{x}\|_\infty := \max_i |x_i|$.

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Example 10

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Example 10

1 ℓ_2 -norm: $\|\mathbf{x}\|_2 := \sqrt{\sum_{i=1}^p x_i^2}$

Euclidean norm

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Euclidean norm

2 ℓ_1 -norm: $\|\mathbf{x}\|_1 := \sum_{i=1}^p |x_i|$

Manhattan norm

Vector Norms

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- (a) $\|\mathbf{x}\| \geq 0$ for all $\mathbf{x} \in \mathbb{R}^p$ non-negativity
- (b) $\|\mathbf{x}\| = 0$ if and only if $\mathbf{x} = \mathbf{0}$ definitiveness
- (c) $\|\lambda \mathbf{x}\| = |\lambda| \|\mathbf{x}\|$ Homogeneity
- (d) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ triangle inequality

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- 2 ℓ_1 -norm: $\|\mathbf{x}\|_1 := \sum_{i=1}^p |x_i|$ Manhattan norm
- 3 ℓ_∞ -norm: $\|\mathbf{x}\|_\infty := \max_{i=1, \dots, p} |x_i|$ Chebyshev norm

Vector norms contd.

Definition 11 (Quasi-norm)

A **quasi-norm** satisfies all the norm properties except (d) triangle inequality, which is replaced by $\|\mathbf{x} + \mathbf{y}\| \leq c (\|\mathbf{x}\| + \|\mathbf{y}\|)$ for a constant $c \geq 1$.

Vector norms contd.

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Definition 12 (Semi(pseudo)-norm)

A **semi(pseudo)-norm** satisfies all the norm properties except (b) definitiveness.

Vector norms contd.

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Definition 13 (ℓ_0 -“norm”)

$$\|\mathbf{x}\|_0 := \lim_{q \downarrow 0} \sum_{i=1}^p |x_i|^q = |\{i : x_i \neq 0\}|$$

Vector norms contd.

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The ℓ_0 -norm counts the non-zero components of $\mathbf{x}$. It is **not** a norm—for nonzero λ , $\|\lambda \mathbf{x}\|_0 = \|\mathbf{x}\|_0$ in general, so homogeneity fails. Hence it is also neither a **quasi-** nor a **semi-norm** under the definitions above.

Vector norms contd.

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Definition 14 (Norm balls)

Radius r in ℓ_q -norm: $\mathcal{B}_q(r) = \{\mathbf{x} \in \mathbb{R}^p : \|\mathbf{x}\|_q \leq r\}$.

Inner products

Definition 15 (Inner product)

The **inner product** of any two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^p$ (denoted by $\langle \cdot, \cdot \rangle$) is defined as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\top \mathbf{y} = \sum_{i=1}^p x_i y_i .$$

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1 $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^p$

symmetry

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- 3 $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ and $\langle \mathbf{x}, \mathbf{x} \rangle = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}, \forall \mathbf{x} \in \mathbb{R}^p$

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Important relations involving the inner product:

Inner products

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- 3 $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ and $\langle \mathbf{x}, \mathbf{x} \rangle = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}, \forall \mathbf{x} \in \mathbb{R}^p$

symmetry

linearity

positive definiteness

Important relations involving the inner product:

- **Hölder's inequality:** $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_q \|\mathbf{y}\|_{q^*}$, where $q, q^* \in [1, \infty]$ and $\frac{1}{q} + \frac{1}{q^*} = 1$ (including $1^* = \infty$).

Inner products

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- **Cauchy–Schwarz** is the special case $q = q^* = 2$ of Hölder's inequality.

Vector norms contd.

Definition 16 (Inner product space)

An **inner product space** is a [vector space](#) endowed with an [inner product](#).

Vector norms contd.

Definition 16 (Inner product space)

An **inner product space** is a **vector space** endowed with an **inner product**.

Definition 17 (Dual norm)

Let $\|\cdot\|$ be a norm on $\mathbb{R}^p$. Its **dual norm**, denoted by $\|\cdot\|^\star$, is

$$\|\mathbf{x}\|^\star = \sup_{\|\mathbf{y}\| \leq 1} \left| \mathbf{x}^\top \mathbf{y} \right|, \quad \text{for all } \mathbf{x} \in \mathbb{R}^p \quad (3)$$

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- 1 ℓ_2 is self-dual: $\sup_{\|\mathbf{x}\|_2 \leq 1} \left| \mathbf{z}^\top \mathbf{x} \right| = \|\mathbf{z}\|_2$.
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Check

The constraint uses the primal norm; the supremum produces its dual. These bounds return in regularization and optimization.

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- A **metric** on a set (of vectors) is a function that satisfies the minimal properties of a distance.

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- A **metric space** $(\mathcal{X}, d)$ is a set $\mathcal{X}$ with a metric d defined on $\mathcal{X}$
- **Norm** induce **metrics** while **pseudo-norms** induce **pseudo-metrics**

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 - Vectors
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 - Two random variables

Basic matrix definitions

Definition 20 (The identity matrix)

The **identity matrix**, denoted $\mathbf{I}_n \in \mathbb{R}^{n \times n}$, is a square matrix with ones on the diagonal and zeros everywhere else. That is,

$$I_{i,j} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases} \quad (4)$$

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Basic matrix definitions contd.

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The **transpose** of a matrix results from “flipping” the rows and columns. Given a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, its transpose, written $\mathbf{A}^\top \in \mathbb{R}^{n \times m}$, is the $n \times m$ matrix whose entries are given by

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Definition 24 (Recall the definition of span)

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 $\Rightarrow$ In that basis case, any vector $\mathbf{v} \in \mathbb{R}^n$ can be written uniquely as a linear combination of $\mathbf{x}_1$ through $\mathbf{x}_k$.
- The **projection** of a vector $\mathbf{y} \in \mathbb{R}^n$ onto the span of $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, is the vector $\mathbf{v} \in \text{span}(\{\mathbf{x}_1, \dots, \mathbf{x}_k\})$ such that $\mathbf{v}$ is as close as possible to $\mathbf{y}$:

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$$\text{Proj}_{\text{range}(\mathbf{A})}(\mathbf{y}) = \arg \min_{\mathbf{v} \in \text{range}(\mathbf{A})} \|\mathbf{v} - \mathbf{y}\|_2 = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{y}, \quad (11)$$

where $\mathbf{A}$ has full column rank, so $p \leq n$.

$\Rightarrow$ This is the fitted vector for least squares. For arbitrary rank, the same unique projection is $\mathbf{AA}^\dagger \mathbf{y}$.

Basic matrix definitions contd.

Definition 26 (Nullspace of a matrix)

The **nullspace** of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, (denoted by $\text{null}(\mathbf{A})$) is defined as

$$\text{null}(\mathbf{A}) = \{\mathbf{x} \in \mathbb{R}^p \mid \mathbf{A}\mathbf{x} = \mathbf{0}\} \quad (12)$$

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- $\text{null}(\mathbf{A})$ is the set of vectors mapped to **zero** by $\mathbf{A}$.
- $\text{null}(\mathbf{A})$ is the set of vectors **orthogonal** to the *rows* of $\mathbf{A}$.

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- $\text{null}(\mathbf{A})$ is the set of vectors mapped to **zero** by $\mathbf{A}$.
- $\text{null}(\mathbf{A})$ is the set of vectors **orthogonal** to the *rows* of $\mathbf{A}$.
- The vectors in $\text{range}(\mathbf{A})$ are in $\mathbb{R}^n$, while the vectors in $\text{null}(\mathbf{A})$ are in $\mathbb{R}^p$.

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 $\Rightarrow$ vectors in $\text{range}(\mathbf{A}^\top)$ and $\text{null}(\mathbf{A})$ are both in $\mathbb{R}^p$.

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 $\Rightarrow$ vectors in $\text{range}(\mathbf{A}^\top)$ and $\text{null}(\mathbf{A})$ are both in $\mathbb{R}^p$.
- $\text{range}(\mathbf{A}^\top)$ and $\text{null}(\mathbf{A})$ intersect only at $\mathbf{0}$ and together span $\mathbb{R}^p$.

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 $\Rightarrow$ vectors in $\text{range}(\mathbf{A}^\top)$ and $\text{null}(\mathbf{A})$ are both in $\mathbb{R}^p$.
- $\text{range}(\mathbf{A}^\top)$ and $\text{null}(\mathbf{A})$ intersect only at $\mathbf{0}$ and together span $\mathbb{R}^p$.
These subspaces are **orthogonal complements**: $\text{range}(\mathbf{A}^\top) = \text{null}(\mathbf{A})^\perp$.

Basic matrix definitions contd.

Definition 27 (Rank of a matrix)

The **rank** of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, (denoted by $\text{rank}(\mathbf{A})$) is defined as

$$\text{rank}(\mathbf{A}) = \dim(\text{range}(\mathbf{A})) \quad (13)$$

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Basic matrix definitions contd.

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- $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$; **and** $\text{rank}(\mathbf{A}) + \dim(\text{null}(\mathbf{A})) = p$.
- For $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times p}$, $\text{rank}(\mathbf{AB}) \leq \min(\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B}))$

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- For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$, $\text{rank}(\mathbf{A} + \mathbf{B}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B})$

Basic matrix definitions contd.

Definition 28 (The inverse)

The **inverse** of a square matrix $A \in \mathbb{R}^{n \times n}$ is denoted A^{-1} , and is the unique matrix such that

Basic matrix definitions contd.

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Basic matrix definitions contd.

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- Not all matrices have inverses. Non-square matrices do not have inverses by definition.

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- The following are properties of the inverse (all assume that $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times n}$ are non-singular):

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 - $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$
 - $(\mathbf{A}^{-1})^\top = (\mathbf{A}^\top)^{-1}$

Basic matrix definitions contd.

Definition 29 (Orthogonal matrices)

- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal** if $\mathbf{x}^\top \mathbf{y} = 0$.

Basic matrix definitions contd.

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- Two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal** if $\mathbf{x}^\top \mathbf{y} = 0$.
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Basic matrix definitions contd.

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- A square matrix $\mathbf{U} \in \mathbb{R}^{n \times n}$ is **orthogonal** if all its columns are orthogonal to each other and are normalized (the columns are then referred to as being **orthonormal**):

$$\mathbf{U}^\top \mathbf{U} = \mathbf{I}_n = \mathbf{U} \mathbf{U}^\top \quad (15)$$

Basic matrix definitions contd.

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Basic matrix definitions contd.

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- The inverse of an orthogonal matrix is its transpose: $\mathbf{U}^{-1} = \mathbf{U}^\top$
- Operating on a vector with an orthogonal matrix will not change its Euclidean norm, i.e.,

$$\|\mathbf{U}\mathbf{x}\|_2 = \|\mathbf{x}\|_2, \quad (16)$$

for any $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{U} \in \mathbb{R}^{n \times n}$ orthogonal.

Basic matrix definitions contd.

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for any $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{U} \in \mathbb{R}^{n \times n}$ orthogonal.

Definition 30 (Symmetric matrix)

A matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **symmetric** if $\mathbf{A} = \mathbf{A}^\top$. It is **skew-symmetric** (or anti-symmetric) if $\mathbf{A} = -\mathbf{A}^\top$.

Basic matrix definitions contd.

Definition 31 (Positive semi-definite & positive definite matrices)

Basic matrix definitions contd.

Definition 31 (Positive semi-definite & positive definite matrices)

A **symmetric** matrix $A \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $A \succeq 0$) if $x^T A x \geq 0$ for all $x \in \mathbb{R}^n$; while it is **positive definite** (denoted $A \succ 0$) if $x^T A x > 0$ for all $x \neq 0$.

Basic matrix definitions contd.

Definition 31 (Positive semi-definite & positive definite matrices)

A **symmetric** matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $\mathbf{A} \succeq 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$; while it is **positive definite** (denoted $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

- $\mathbf{A} \succeq 0$ iff all its **eigenvalues** are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.

Basic matrix definitions contd.

Definition 31 (Positive semi-definite & positive definite matrices)

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- $\mathbf{A} \succeq 0$ iff all its **eigenvalues** are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.
- Similarly, $\mathbf{A} \succ 0$ iff all its **eigenvalues** are **positive**, i.e., $\lambda_{\min}(\mathbf{A}) > 0$.

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- $\mathbf{A} \succeq 0$ iff all its **eigenvalues** are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.
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- $\mathbf{A}$ is **negative semi-definite** if $-\mathbf{A} \succeq 0$; while $\mathbf{A}$ is **negative definite** if $-\mathbf{A} \succ 0$.

Basic matrix definitions contd.

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A **symmetric** matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $\mathbf{A} \succeq 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$; while it is **positive definite** (denoted $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

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- $\mathbf{A}$ is **negative semi-definite** if $-\mathbf{A} \succeq 0$; while $\mathbf{A}$ is **negative definite** if $-\mathbf{A} \succ 0$.
- **Semi-definite ordering** of two *symmetric* matrices, $\mathbf{A}$ and $\mathbf{B}$: $\mathbf{A} \succeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B} \succeq 0$.

Basic matrix definitions contd.

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A **symmetric** matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $\mathbf{A} \succeq 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$; while it is **positive definite** (denoted $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

- $\mathbf{A} \succeq 0$ iff all its **eigenvalues** are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.
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Example 32 (Matrix inequalities)

Basic matrix definitions contd.

Definition 31 (Positive semi-definite & positive definite matrices)

A **symmetric** matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $\mathbf{A} \succeq 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$; while it is **positive definite** (denoted $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

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- **Semi-definite ordering** of two *symmetric* matrices, $\mathbf{A}$ and $\mathbf{B}$: $\mathbf{A} \succeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B} \succeq 0$.

Example 32 (Matrix inequalities)

- 1 If $\mathbf{A} \succeq 0$ and $\mathbf{B} \succeq 0$, then $\mathbf{A} + \mathbf{B} \succeq 0$

Basic matrix definitions contd.

Definition 31 (Positive semi-definite & positive definite matrices)

A **symmetric** matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $\mathbf{A} \succeq 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$; while it is **positive definite** (denoted $\mathbf{A} \succ 0$) if $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

- $\mathbf{A} \succeq 0$ iff all its **eigenvalues** are **non-negative**, i.e., $\lambda_{\min}(\mathbf{A}) \geq 0$.
- Similarly, $\mathbf{A} \succ 0$ iff all its **eigenvalues** are **positive**, i.e., $\lambda_{\min}(\mathbf{A}) > 0$.
- $\mathbf{A}$ is **negative semi-definite** if $-\mathbf{A} \succeq 0$; while $\mathbf{A}$ is **negative definite** if $-\mathbf{A} \succ 0$.
- **Semi-definite ordering** of two *symmetric* matrices, $\mathbf{A}$ and $\mathbf{B}$: $\mathbf{A} \succeq \mathbf{B}$ if $\mathbf{A} - \mathbf{B} \succeq 0$.

Example 32 (Matrix inequalities)

- 1 If $\mathbf{A} \succeq 0$ and $\mathbf{B} \succeq 0$, then $\mathbf{A} + \mathbf{B} \succeq 0$
- 2 If $\mathbf{A} \succeq \mathbf{B}$ and $\mathbf{C} \succeq \mathbf{D}$, then $\mathbf{A} + \mathbf{C} \succeq \mathbf{B} + \mathbf{D}$.

Basic matrix definitions contd.

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Basic matrix definitions contd.

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A **symmetric** matrix $A \in \mathbb{R}^{n \times n}$ is **positive semi-definite** (denoted $A \succeq 0$) if $x^T A x \geq 0$ for all $x \in \mathbb{R}^n$; while it is **positive definite** (denoted $A \succ 0$) if $x^T A x > 0$ for all $x \neq 0$.

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- 2 If $A \succeq B$ and $C \succeq D$, then $A + C \succeq B + D$.
- 3 If $B \preceq 0$, then $A + B \preceq A$
- 4 If $A \succeq 0$ and $\alpha \geq 0$, then $\alpha A \succeq 0$
- 5 If $A \succ 0$, then $A^2 \succ 0$

Basic matrix definitions contd.

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- 4 If $A \succeq 0$ and $\alpha \geq 0$, then $\alpha A \succeq 0$
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Matrix decompositions

Definition 33 (Eigenvalues & Eigenvectors)

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A nonzero vector $\mathbf{x}$ is an **eigenvector** of a *square* matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ if $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$. For a general real matrix, λ and $\mathbf{x}$ may be complex; for real symmetric $\mathbf{A}$, they can be chosen real.

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- *The trace of $\mathbf{A}$ is equal to the sum of its eigenvalues, $\text{trace}(\mathbf{A}) = \sum_{i=1}^n \lambda_i$*

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- The rank of $\mathbf{A}$ is equal to the number of non-zero eigenvalues of $\mathbf{A}$
- If $\mathbf{A}$ is non-singular, then $1/\lambda_i$ is an eigenvalue of $\mathbf{A}^{-1}$ with associated eigenvector $\mathbf{x}_i$, i.e., $\mathbf{A}^{-1}\mathbf{x}_i = (1/\lambda_i)\mathbf{x}_i$

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- The eigenvalues of a diagonal matrix $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ are just the diagonal entries $d_1, \dots, d_n$

Matrix decompositions contd.

Definition 35 (Eigenvalue decomposition)

If a **square** matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ has a basis of real eigenvectors, its **eigenvalue decomposition** is

$$\mathbf{A} = \mathbf{X}\mathbf{\Lambda}\mathbf{X}^{-1} \tag{17}$$

Matrix decompositions contd.

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Definition 36 (Eigendecomposition of symmetric matrices)

If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is **symmetric**, the decomposition becomes $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ is **orthogonal**, i.e., $\mathbf{U}^T \mathbf{U} = \mathbf{I}_n$ and λ_i are real.

Matrix decompositions contd.

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If we order $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, $\lambda_i(\mathbf{A})$ becomes the i^{th} largest eigenvalue of $\mathbf{A}$:

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- $\lambda_n(\mathbf{A}) = \lambda_{\min}(\mathbf{A})$ is the minimum eigenvalue of $\mathbf{A}$
- $\lambda_1(\mathbf{A}) = \lambda_{\max}(\mathbf{A})$ is the maximum eigenvalue of $\mathbf{A}$

Matrix decompositions contd

Definition 37 (Singular value decomposition)

The **singular value decomposition** (SVD) of a matrix, $\mathbf{A} \in \mathbb{R}^{n \times p}$, is given by:

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\top} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^{\top} \quad (18)$$

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- $\mathbf{U} \in \mathbb{R}^{n \times r}$ and $\mathbf{V} \in \mathbb{R}^{p \times r}$ have orthonormal columns: $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_r = \mathbf{V}^\top \mathbf{V}$

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- $\mathbf{\Sigma} = \text{diag}(\sigma_1, \dots, \sigma_r)$, where $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ (for $r = 0$, the sum is empty)

Matrix decompositions contd

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- $\mathbf{v}_i$ are **eigenvectors** of $\mathbf{A}^\top \mathbf{A}$, $\sigma_i = \sqrt{\lambda_i(\mathbf{A}^\top \mathbf{A})}$, and $\lambda_i(\mathbf{A}^\top \mathbf{A}) = 0$ for $i > r$, since $\mathbf{A}^\top \mathbf{A} = (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top) = (\mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top)$

Matrix decompositions contd

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Matrix decompositions contd

Definition 38 (LU)

With row pivoting, the **LU factorization** of a **non-singular square** matrix, $\mathbf{A} \in \mathbb{R}^{p \times p}$, is given by:

$$\mathbf{PA} = \mathbf{LU}, \quad (19)$$

where $\mathbf{P}$ is a **permutation matrix**^a, $\mathbf{L}$ is **lower triangular** and $\mathbf{U}$ is **upper triangular**.

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Matrix decompositions contd

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Definition 40 (Cholesky)

The **Cholesky factorization** of a **positive definite and symmetric** matrix, $\mathbf{A} \in \mathbb{R}^{p \times p}$, is given by $\mathbf{A} = \mathbf{LL}^T$, where $\mathbf{L}$ is a **lower triangular** matrix with **positive** entries on the *diagonal*.

Matrix norms

Definition 41 (Matrix norm)

A norm of an $n \times p$ matrix is a map $\|\cdot\| : \mathbb{R}^{n \times p} \rightarrow \mathbb{R}$ such that for all matrices $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n \times p}$ and scalar $\lambda \in \mathbb{R}$

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Definition 44 (Operator norm)

For $\mathbf{A} \in \mathbb{R}^{n \times p}$, $q, r \in [1, \infty]$, and $\mathbf{x} \in \mathbb{R}^p$ (so $\mathbf{Ax} \in \mathbb{R}^n$),

$$\|\mathbf{A}\|_{q \rightarrow r} := \sup_{\|\mathbf{x}\|_q \leq 1} \|\mathbf{Ax}\|_r.$$

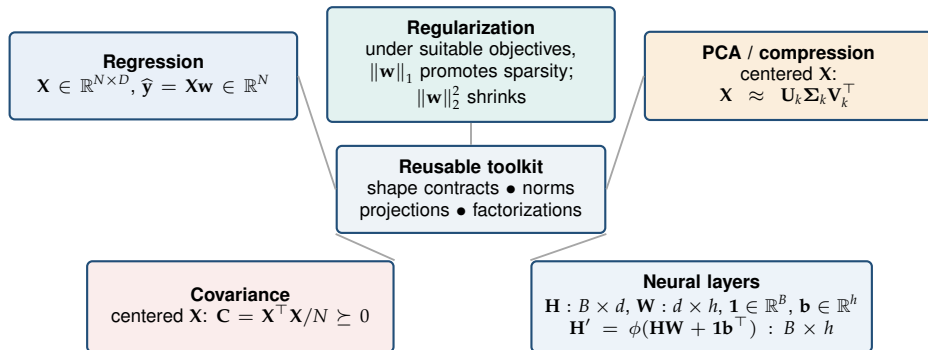
In particular, $\|\mathbf{A}\|_{2 \rightarrow 2} = \sigma_1(\mathbf{A})$ is the largest Euclidean stretch, with $\sigma_1(\mathbf{0}) := 0$ for the zero matrix.

Where linear algebra returns in this course

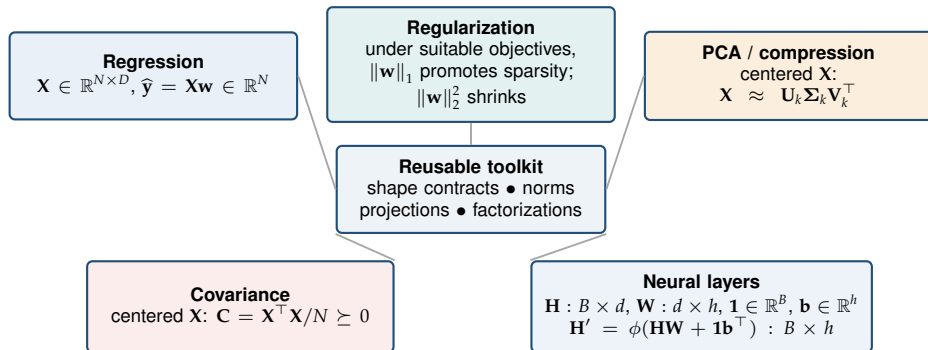
Reusable toolkit

shape contracts • norms
projections • factorizations

Where linear algebra returns in this course



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The reusable habit

Name every object's space, check every inner dimension, then ask what geometry the operation preserves or discards.

Calculus recap 1/2: Gradients

Recall before the bridge

For a differentiable $f : \mathbb{R}^D \rightarrow \mathbb{R}$, which unit direction gives the largest local decrease, and what qualification belongs in that claim?

Commit to an answer before advancing the slide.

Calculus recap 1/2: Gradients

Bridge to later lectures: notation we will reuse in optimization and neural networks.

1. Recall the sensitivities

In one variable,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

For $\mathbf{x} = (x_1, \dots, x_D)^\top$ and coordinate vector $\mathbf{e}_i$,

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\mathbf{e}_i) - f(\mathbf{x})}{h}.$$

The course uses a column gradient:

$$\nabla f(\mathbf{x}) = [\partial f / \partial x_1 \quad \dots \quad \partial f / \partial x_D]^\top \in \mathbb{R}^{D \times 1}.$$

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2. Use the local linear model

If f is differentiable at $\mathbf{x}$, then

$$f(\mathbf{x} + \Delta \mathbf{x}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \Delta \mathbf{x} + o(\|\Delta \mathbf{x}\|_2).$$

Thus for $\|\mathbf{d}\|_2 = 1$, the directional derivative is

$$D_{\mathbf{d}}f(\mathbf{x}) = \nabla f(\mathbf{x})^\top \mathbf{d}, \quad (1 \times D)(D \times 1) = 1 \times 1.$$

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3. Result and quick check

Cauchy–Schwarz gives $|\nabla f(\mathbf{x})^\top \mathbf{d}| \leq \|\nabla f(\mathbf{x})\|_2$. If $\nabla f(\mathbf{x}) \neq \mathbf{0}$, the locally steepest Euclidean descent direction is

$$\mathbf{d}_- = - \frac{\nabla f(\mathbf{x})}{\|\nabla f(\mathbf{x})\|_2}.$$

This is a **first-order, local** statement; a finite update still needs a step size. If $\nabla f(\mathbf{x}) = \mathbf{0}$, first order selects no direction.

Calculus recap 2/2: Chain rule and backpropagation

Shape check before the bridge

Let $g : \mathbb{R}^D \rightarrow \mathbb{R}^K$ and $f : \mathbb{R}^K \rightarrow \mathbb{R}$. Which product maps the upstream K -gradient back to a gradient in $\mathbb{R}^D$?

Commit to an answer before advancing the slide.

Calculus recap 2/2: Chain rule and backpropagation

Bridge to later lectures: composition notation for the later backpropagation lecture.

1. Name the local map and its shape

The scalar rule is

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

For $\mathbf{g} : \mathbb{R}^D \rightarrow \mathbb{R}^K$, define

$$[\mathbf{J}_{\mathbf{g}}(\mathbf{x})]_{ki} = \frac{\partial g_k}{\partial x_i}(\mathbf{x}), \quad \mathbf{J}_{\mathbf{g}}(\mathbf{x}) \in \mathbb{R}^{K \times D}.$$

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2. Compose compatible shapes

With column gradients,

$$\nabla_{\mathbf{x}}(f \circ \mathbf{g})(\mathbf{x}) = \mathbf{J}_{\mathbf{g}}(\mathbf{x})^{\top} \nabla f(\mathbf{g}(\mathbf{x})).$$

The check is

$$(D \times K)(K \times 1) = D \times 1.$$

Hence $\mathbf{J}_{\mathbf{g}} \nabla f$ is not the chain-rule product under these conventions.

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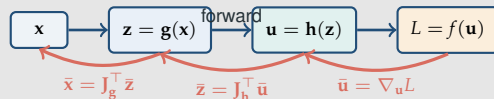
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3. Preview reverse-mode backpropagation



For scalar L , $\bar{\mathbf{v}} := \nabla_{\mathbf{v}} L$ has the same dimension as $\mathbf{v}$. Backpropagation stores or recomputes needed forward intermediates, then propagates local Jacobian-transpose-vector products in reverse and sums contributions at branches. **Later:** optimization will decide how to use the resulting gradients.

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- $P(A|B)$ is the probability measure of the event A after observing the occurrence of B .
- Two events are called independent if and only if $P(A \cap B) = P(A)P(B)$.
- **Diagnostic:** Are disjoint positive-probability events independent?

Properties:

- If $A \subseteq B \Rightarrow P(A) \leq P(B)$
- $P(A \cap B) \leq \min(P(A), P(B))$
- (Union Bound) $P(A \cup B) \leq P(A) + P(B)$
- $P(\Omega \setminus A) = 1 - P(A)$
- If $A_1, \dots, A_k$ partition Ω , then $\sum_{i=1}^k P(A_i) = 1$ and, for any event B ,

$$P(B) = \sum_{i: P(A_i) > 0} P(B \mid A_i) P(A_i) \quad (\text{law of total probability}).$$

Definition 48 (Conditional probability and independence)

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- **Diagnostic:** Are disjoint positive-probability events independent?
- **Answer:** No—they are dependent.

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 - Two random variables

Random Variables

Definition 49 (Random Variable)

A real-valued random variable, $X(\omega)$ or X , is a **function** $X : \Omega \rightarrow \mathbb{R}$ that associates a value with each outcome of a randomized experiment.

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Remark 51

- *A random variable must make value statements into legitimate events.*
- *Concretely, for each interval $C \subseteq \mathbb{R}$ used here, $\{\omega : X(\omega) \in C\}$ must belong to $\mathcal{F}$.*

Example 52 (Discrete random variable)

The probability of the set associated with a random variable X taking on some specific value k is

$$P(X = k) \triangleq P(\{\omega : X(\omega) = k\}) . \quad (22)$$

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The probability that X takes on a value between two real constants a and b (where $a < b$) as

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Cumulative distribution functions

The CDF records the distribution of a real-valued random variable.

Definition 54 (CDF)

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- F_X is right-continuous, and $P(a < X \leq b) = F_X(b) - F_X(a)$.

Probability mass functions

Let a discrete random variable (RV) X take values in a finite or countable support $\mathcal{X}$.

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A **Probability Mass Function** (PMF) is a function $p_X : \mathcal{X} \rightarrow [0, 1]$ such that

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For a continuous RV with a density, probabilities are areas under a non-negative function.

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A **Probability Density Function** (PDF) $f_X : \mathbb{R} \rightarrow [0, \infty)$ satisfies

$$F_X(x) = \int_{-\infty}^x f_X(t) dt. \quad (26)$$

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- $\int_{\mathcal{A}} f_X(x) dx = P(X \in \mathcal{A})$ for suitable value sets $\mathcal{A}$.
- For example, $X \sim \text{Uniform}(0, \frac{1}{2})$ has density height 2 but total area 1.

Expectation

Definition 57 (Expectation of random variable)

Discrete random variable: Suppose that X has support $\mathcal{X}$, PMF p_X , and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a function for which the expectation exists.

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- **Diagnostic:** Does linearity require independence?
- **Answer:** No.

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If X has a finite second moment, its variance is

$$\text{Var}(X) \triangleq \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2. \quad (30)$$

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$$p_X(x) = \begin{cases} q & \text{if } x = 1 \\ 1 - q & \text{if } x = 0 \end{cases} . \quad (31)$$

- $X \sim \text{Binomial}(n, q)$ (where $0 \leq q \leq 1$): the number of heads in n independent flips of a coin with heads probability q

$$p_X(x) = \binom{n}{x} q^x (1 - q)^{n-x}, \quad x \in \{0, \dots, n\} . \quad (32)$$

- $X \sim \text{Geometric}(q)$ (where $0 < q \leq 1$): the number of flips of a coin with heads probability q until the first heads.

$$p_X(x) = q(1 - q)^{x-1}, \quad x \in \{1, 2, \dots\} . \quad (33)$$

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$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} . \quad (37)$$

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Joint and marginal CDFs

Definition 59 (Joint CDF $F_{XY}(x, y)$)

The joint cumulative distribution function of X and Y is defined by

$$F_{XY}(x, y) = P(X \leq x, Y \leq y) . \quad (38)$$

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Definition 60 (Marginal CDFs)

The joint CDF and the CDFs of each variable separately are related by

$$F_X(x) = \lim_{y \rightarrow \infty} F_{XY}(x, y), \quad F_Y(y) = \lim_{x \rightarrow \infty} F_{XY}(x, y). \quad (39)$$

where F_X and F_Y are the **marginal cumulative distribution functions**.

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- If either argument tends to $-\infty$, then $F_{XY}(x, y) \rightarrow 0$.

Joint and marginal Probability Mass Functions (PMFs)

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If X and Y are discrete random variables with supports $\mathcal{X}$ and $\mathcal{Y}$, then the **joint PMF** $p_{XY} : \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]$ is

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Definition 62 (Marginal PMF)

We refer to $p_X(x)$ as the **marginal PMF** of X .

Joint and marginal Probability Density Functions (PDFs)

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Let X and Y be two continuous random variables with joint distribution function F_{XY} . When their joint law has a density and F_{XY} is twice differentiable, the **joint PDF** is

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Definition 64 (Marginal PDF)

The marginal PDF of X is $f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$.

Conditional distributions

Definition 65 (Conditional distribution)

Discrete case: for x with $p_X(x) > 0$, the conditional PMF of Y given $X = x$ is

$$p_{Y|X}(y|x) = \frac{p_{XY}(x, y)}{p_X(x)}, \quad (42)$$

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In either ratio, the denominator must be strictly positive.

Bayes's rule

Diagnostic: What denominator conditions must be checked before applying Bayes's rule?

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Definition 66 (Bayes's rule)

For **discrete random variables** X, Y , at values with $p_X(x) > 0$ (and $p_Y(y) > 0$ for the likelihood form),

$$p_{Y|X}(y|x) = \frac{p_{XY}(x, y)}{p_X(x)} = \frac{p_{X|Y}(x|y)p_Y(y)}{\sum_{y' \in \mathcal{Y}} p_{XY}(x, y')} . \quad (44)$$

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- $P(B|A)$ is the probability of observing B given A . As a function of A with B fixed, this is the **likelihood**.

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Lemma 67

If X and Y are independent, then for any value sets $\mathcal{A}$, $\mathcal{B} \subseteq \mathbb{R}$ under consideration,

$$P(X \in \mathcal{A}, Y \in \mathcal{B}) = P(X \in \mathcal{A})P(Y \in \mathcal{B}) \quad (46)$$

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Two random variables X and Y are **independent** if $F_{XY}(x, y) = F_X(x)F_Y(y)$ for all values of x and y .

- For discrete random variables, $p_{XY}(x, y) = p_X(x)p_Y(y)$ for all $x \in \text{Val}(X), y \in \text{Val}(Y)$.
- For discrete random variables, $p_{Y|X}(y|x) = p_Y(y)$ whenever $p_X(x) > 0$, for all $y \in \text{Val}(Y)$.
- For continuous random variables, $f_{XY}(x, y) = f_X(x)f_Y(y)$ (up to equality almost everywhere).
- For continuous random variables, $f_{Y|X}(y|x) = f_Y(y)$ whenever $f_X(x) > 0$, for all $y \in \mathbb{R}$.

Lemma 67

If X and Y are independent, then for any value sets $\mathcal{A}, \mathcal{B} \subseteq \mathbb{R}$ under consideration,

$$P(X \in \mathcal{A}, Y \in \mathcal{B}) = P(X \in \mathcal{A})P(Y \in \mathcal{B}) \quad (46)$$

Remark 68

If X is independent of Y , then any function of X is independent of any function of Y .

Expectation and covariance

Definition 69 (Expectation)

Discrete case: Suppose that X, Y are discrete and $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a function for which the expectation exists. The expected value of $g(X, Y)$ is defined as:

$$\mathbb{E}[g(X, Y)] \triangleq \sum_{x \in \text{Val}(X)} \sum_{y \in \text{Val}(Y)} g(x, y) p_{XY}(x, y) . \quad (47)$$

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Continuous case: For continuous random variables X, Y , the analogous expression is

$$\mathbb{E}[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y) dy dx. \quad (48)$$

Covariance

Definition 70 (Covariance)

If X and Y have finite second moments, their covariance is

$$\text{Cov}(X, Y) \triangleq \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] . \quad (49)$$

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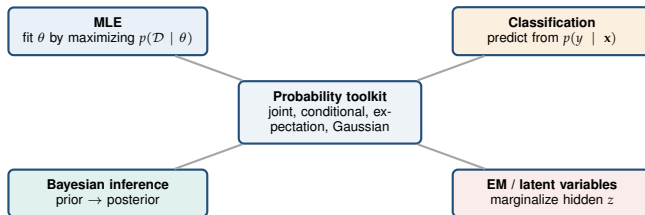
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- **Counterexample:** If $X \sim \text{Uniform}(-1, 1)$ and $Y = X^2$, they are dependent but uncorrelated.

Where probability returns in this course

Probability toolkit
joint, conditional, expectation, Gaussian

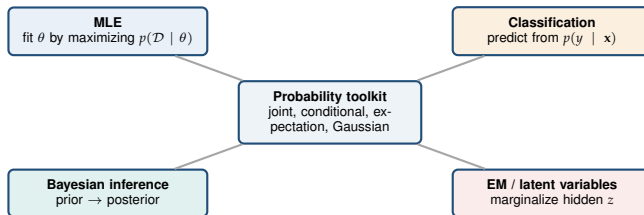
Goodfellow–Bengio–Courville, 2016, §§3, 5.5, 19.2.

Where probability returns in this course



Goodfellow–Bengio–Courville, 2016, §§3, 5.5, 19.2.

Where probability returns in this course



Carry-forward checklist

Mass, density, or probability? Positive conditioning denominator?
Which variable is marginalized? Which support/moment assumptions apply?

Goodfellow–Bengio–Courville, 2016, §§3, 5.5, 19.2.